

Powers of cycles have equivalence covering number $\Theta(\log k)$: a negative answer to an OpenProblemGarden conjecture

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Machine-generated note

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Abstract

For $k \geq 1$ and $n \geq 2k + 2$, we prove

$$1 + \lceil \log_2(k + 1) \rceil \leq \text{eq}(C_n^k) \leq 1 + 2 \lceil \log_2(k + 1) \rceil,$$

with equality in the lower bound whenever $k + 1$ divides n . Thus the equivalence covering number of noncomplete powers of cycles is $\Theta(\log(k + 1))$, uniformly in n , disproving the proposed $\Omega(k)$ lower bound even when $n/k \rightarrow \infty$.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

An *equivalence graph* is a disjoint union of cliques. An equivalence cover of a graph G is a family of equivalence subgraphs whose edge sets cover $E(G)$, and $\text{eq}(G)$ is the minimum size of such a family. We allow isolated vertices, so every member of a cover may be regarded as spanning. The graph C_n^k has vertex set $\mathbb{Z}/n\mathbb{Z}$, with two vertices adjacent when their cyclic distance is at most k .

The OpenProblemGarden entry [1], originating in a problem page of West [2], asks verbatim:

Conjecture 1.1 (OpenProblemGarden). Given k and n , the graph C_n^k has equivalence covering number $\Omega(k)$.

The discussion emphasizes the regime in which n is large compared with k . Equivalence coverings go back at least to Alon [3]; Esperet, Gimbel and King studied the related parameter for line graphs [4]. The lower bound used here is a matrix-rank version of Alon's Theorem 1.1, and the case $n = 2k + 2$ of our exact formula is already Alon's Corollary 1.2. The contribution of the present argument is the binary upper-bound construction for all n , especially the divisible cases with three or more blocks and hence $n \gg k$.

The source page [2] states that $\text{eq}(C_n^k) = k + 1$ when $k + 1 \mid n$. That assertion is false: Alon's corollary already gives $\text{eq}(C_{2k+2}^k) = \lceil \log_2(2k + 2) \rceil$, and the construction below extends this phenomenon to every multiple $n = m(k + 1)$ with $m \geq 2$. Work on the edge clique-cover number of cycle powers [5] concerns a different parameter.

2 Main result

Theorem 2.1 (Uniform bounds and an exact divisible case). *Let $k \geq 1$, set $s = k + 1$ and $r = \lceil \log_2 s \rceil$. For every $n \geq 2s$,*

$$r + 1 \leq \text{eq}(C_n^k) \leq 2r + 1.$$

If $s \mid n$, then $\text{eq}(C_n^k) = r + 1$. For $3 \leq n \leq 2k + 1$, one has $C_n^k = K_n$ and $\text{eq}(C_n^k) = 1$.

Idea of proof. Divide the cycle into blocks of s consecutive vertices. One layer covers edges inside blocks. At each of r binary levels, vertices whose first differing bit is 1 on the left and 0 on the right form disjoint cliques across every block boundary. This gives $r + 1$ layers when the blocks close cyclically and, after treating the final cut separately, $2r + 1$ layers for arbitrary n . Alon’s rank obstruction, applied to two sequences of $2s$ vertices at cyclic offset s , yields $2s \leq 2^{\text{eq}(C_n^k)}$. [\[full proof\]](#)

Corollary 2.2. *The conjecture in [Conjecture 1.1](#) is false, even along families with $n/k \rightarrow \infty$.*

Idea of proof. Take $k = 2^q - 1$ and $n = 2^{2^q}$. Then $k + 1 \mid n$ and [Theorem 2.1](#) gives $\text{eq}(C_n^k) = q + 1$, whereas $(q + 1)/(2^q - 1) \rightarrow 0$ and $n/k \rightarrow \infty$. [\[full proof\]](#)

3 Binary layers and Alon’s obstruction

Write every $x \in \{0, \dots, s - 1\}$ with r binary digits, most significant first. For a position h , a prefix p of length $h - 1$, and $\varepsilon \in \{0, 1\}$, put

$$I_h^\varepsilon(p) = \{x : (x_1, \dots, x_{h-1}) = p, x_h = \varepsilon\}.$$

Every element of $I_h^1(p)$ is larger than every element of $I_h^0(p)$; conversely, if $x > y$, their first differing bit supplies such h, p .

Lemma 3.1 (Binary boundary cover). *If $n = ms$ with $m \geq 2$, then C_n^{s-1} has an equivalence cover with $r + 1$ layers. For arbitrary $n \geq 2s$, it has an equivalence cover with at most $2r + 1$ layers.*

Idea of proof. For $n = ms$, use the block cliques and, at level h , join the 1-half of each prefix in block i to its 0-half in block $i + 1$. A vertex chooses exactly one of its two adjacent boundaries at each level, so the resulting cliques are disjoint. For arbitrary n , the same construction on a line covers all non-wrapping edges and r additional levels cover the wrap-around cut. [\[full proof\]](#)

Lemma 3.2 (Alon’s rank obstruction). *For $n \geq 2s$,*

$$\text{eq}(C_n^{s-1}) \geq \lceil \log_2(2s) \rceil = 1 + \lceil \log_2 s \rceil.$$

Idea of proof. Encode each cover layer by component labels and multiply their pairwise differences. The resulting matrix is a sum of at most 2^t rank-one matrices, where t is the number of layers. Its submatrix with rows $0, \dots, 2s - 1$ and columns shifted by s is triangular with nonzero diagonal, so it has rank $2s$. This is the specialization of Alon’s Theorem 1.1 used here. [\[full proof\]](#)

4 Remarks

Remark 4.1 (Nondivisible cases). The exact value is not determined when $k + 1 \nmid n$. The referee’s exhaustive search found, for example, $\text{eq}(C_7^2) = 4 > 3$ and $\text{eq}(C_9^3) = \text{eq}(C_{10}^3) = \text{eq}(C_{11}^3) = 4 > 3$, so the lower bound need not be tight. The upper bound $2r + 1$ is not claimed to be tight.

Remark 4.2 (Conventions). The covering cliques need not be induced, maximal, or consecutive. This is the standard definition in [\[3, 4, 1\]](#). Some binary sets are empty or singletons when s is not a power of two; deleting empty sets and retaining or deleting singletons does not affect the edge cover. The case $k = 0$ is outside the theorem and has $\text{eq}(C_n^0) = 0$.

Remark 4.3 (Computational audit). The referee independently checked the constructions for 7,390 pairs with $k \leq 24$ and $n \leq 300$, with no failure. Exact set-cover computations gave

$$\text{eq}(C_8^3) = \text{eq}(C_{12}^3) = 3, \quad \text{eq}(C_{10}^4) = 4,$$

contradicting the $k + 1$ assertion on the source page. It also checked the matrix zero pattern, the triangular $2s \times 2s$ minor and the rank inequalities for all tested $m \in \{2, 3\}$ and $k \leq 15$.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by gpt-6-astra, reasoning effort max, in a single pass started on 2026-09-16, for OpenProblemGarden catalog id covering_powers_of_cycles_with_equivalence_subgraphs, campaign leg attacks_opg. (2) An independent adversarial referee agent (claude-opus-5) checked every step, compared the claim with the source literature, and ran the computations described above on 2026-09-17; its verdict was CONFIRMED. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-17 claude-fable-5-1 began the rewrite but left only a compilable skeleton containing the preamble, title and section headings. (4) On 2026-09-18 Codex completed the present note. It reorganised the original argument with full proofs in [Appendix A](#), attributed the lower bound and the $n = 2k + 2$ case to Alon, explained the erroneous assertion on West’s source page, added the related-work citations and computational audit, and made explicit why deleting out-of-range indices in a short final block preserves the cover. No other mathematical content was changed. **No human mathematician has checked the argument.**

A Full proofs

Lemma 3.1 (Binary boundary cover). *If $n = ms$ with $m \geq 2$, then C_n^{s-1} has an equivalence cover with $r + 1$ layers. For arbitrary $n \geq 2s$, it has an equivalence cover with at most $2r + 1$ layers.*

Proof. First suppose $n = ms$. Partition the cyclic order into blocks

$$B_i = \{v_{i,0}, \dots, v_{i,s-1}\}, \quad i \in \mathbb{Z}/m\mathbb{Z}.$$

Let H_0 be the union of the cliques on the B_i . For $1 \leq h \leq r$ and every prefix p of length $h - 1$, define

$$Q_{i,h,p} = \{v_{i,x} : x \in I_h^1(p)\} \cup \{v_{i+1,y} : y \in I_h^0(p)\},$$

and let H_h be the union of the nonempty cliques $Q_{i,h,p}$.

Each $Q_{i,h,p}$ is a clique. Pairs within one block have cyclic distance at most $s - 1$. For a cross-block pair, $x > y$ and the clockwise distance is $s + y - x \leq s - 1$. For a fixed h , the cliques are vertex-disjoint: if $x_h = 1$ then $v_{i,x}$ occurs only in $Q_{i,h,p}$, while if $x_h = 0$ it occurs only in $Q_{i-1,h,p}$, with p its unique prefix.

Every edge is covered. A short cyclic arc of length at most $s - 1$ crosses at most one block boundary. Edges inside a block lie in H_0 . Across the boundary from B_i to B_{i+1} , adjacency of $v_{i,x}$ and $v_{i+1,y}$ gives $s + y - x \leq s - 1$, hence $x > y$; their first differing bit places them together in some $Q_{i,h,p}$. Thus $H_0, H_1, \dots, H_r$ form a cover.

For arbitrary $n \geq 2s$, first label the vertices linearly by $0, \dots, n - 1$ and divide them into consecutive blocks of size s , allowing the last block to be shorter. Apply the same construction across consecutive block boundaries. More explicitly, define the construction as though the last block had s local indices and then delete every out-of-range vertex. Intersecting each clique with the actual vertex set preserves cliqueness and pairwise disjointness. An actual edge either lies in one block or crosses one boundary; in the latter case both of its actual endpoints remain, so its first-bit witness remains as well. Hence $r + 1$ layers cover all edges whose ordinary index difference is at most $s - 1$.

It remains to cover wrapping edges. Put

$$A = \{n - s, \dots, n - 1\}, \quad B = \{0, \dots, s - 1\};$$

these sets are disjoint cliques because $n \geq 2s$. Write $a_x = n - s + x$ and $b_y = y$. The pair $a_x b_y$ is a wrapping edge exactly when $n - a_x + b_y = s + y - x \leq s - 1$, equivalently $x > y$. At every binary level use the cliques

$$\{a_x : x \in I_h^1(p)\} \cup \{b_y : y \in I_h^0(p)\}.$$

For fixed h these are disjoint, and the first differing bit covers every wrapping edge. These r layers together with the $r + 1$ linear layers give $2r + 1$ layers. $\square$

Lemma 3.2 (Alon’s rank obstruction). *For $n \geq 2s$,*

$$\text{eq}(C_n^{s-1}) \geq \lceil \log_2(2s) \rceil = 1 + \lceil \log_2 s \rceil.$$

Proof. Let $t = \text{eq}(C_n^{s-1})$. Make every cover layer spanning by adding isolated vertices. In layer ℓ , assign distinct real labels to its clique components and let $c_\ell(v)$ be the label of the component containing v . Because each component is a clique and every edge is covered, for distinct vertices u, v ,

$$uv \in E(C_n^{s-1}) \iff c_\ell(u) = c_\ell(v) \text{ for some } \ell.$$

Define

$$M_{uv} = \prod_{\ell=1}^t (c_\ell(u) - c_\ell(v)).$$

Thus $M_{uv} = 0$ exactly when $u = v$ or uv is an edge. For $S \subseteq [t]$ put $a_S(v) = \prod_{\ell \in S} c_\ell(v)$. Expanding gives

$$M = \sum_{S \subseteq [t]} (-1)^{t-|S|} a_S a_{[t] \setminus S}^T,$$

a sum of 2^t rank-one matrices, so $\text{rank } M \leq 2^t$.

With vertex labels modulo n , take the submatrix

$$N_{ij} = M_{i, (s+j) \bmod n}, \quad 0 \leq i, j < 2s.$$

Both index lists are injective because $2s \leq n$. If $i > j$, then $1-s \leq s+j-i \leq s-1$, so the corresponding vertices are equal or adjacent and $N_{ij} = 0$. On the diagonal their cyclic distance is s (also when $n = 2s$), so they are distinct and nonadjacent and $N_{ii} \neq 0$. Therefore N is upper triangular with nonzero diagonal. Hence $2s = \text{rank } N \leq \text{rank } M \leq 2^t$, which implies $t \geq \lceil \log_2(2s) \rceil = 1 + \lceil \log_2 s \rceil$. $\square$

Theorem 2.1 (Uniform bounds and an exact divisible case). *Let $k \geq 1$, set $s = k + 1$ and $r = \lceil \log_2 s \rceil$. For every $n \geq 2s$,*

$$r + 1 \leq \text{eq}(C_n^k) \leq 2r + 1.$$

If $s \mid n$, then $\text{eq}(C_n^k) = r + 1$. For $3 \leq n \leq 2k + 1$, one has $C_n^k = K_n$ and $\text{eq}(C_n^k) = 1$.

Proof. The upper bounds are [Lemma 3.1](#), and the lower bound is [Lemma 3.2](#). When $s \mid n$, the two bounds $\text{eq}(C_n^{s-1}) \leq r + 1$ and $\text{eq}(C_n^{s-1}) \geq r + 1$ coincide. Finally, if $3 \leq n \leq 2k + 1$, the largest cyclic distance is $\lfloor n/2 \rfloor \leq k$, so $C_n^k = K_n$, which is covered by one clique. $\square$

Corollary 2.2. *The conjecture in [Conjecture 1.1](#) is false, even along families with $n/k \rightarrow \infty$.*

Proof. For $q \geq 1$, take $k = 2^q - 1$ and $n = 2^{2q}$. Then $s = 2^q$ divides n and [Theorem 2.1](#) gives $\text{eq}(C_n^k) = q + 1$. Meanwhile

$$\frac{\text{eq}(C_n^k)}{k} = \frac{q + 1}{2^q - 1} \rightarrow 0, \quad \frac{n}{k} = \frac{2^{2q}}{2^q - 1} \rightarrow \infty.$$

$\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/covering_powers_of_cycles_with_equivalence_subgraphs/`. Treat it as machine output, not as an authority.

B.1 Interpretation

Definitions used. The writeup uses: an *equivalence subgraph* of G is a spanning subgraph whose connected components are cliques of G ; $\text{eq}(G)$ is the least number of such subgraphs whose edge sets cover $E(G)$. This matches the OpenProblemGarden entry verbatim (“a subgraph H of G is an equivalence subgraph of G if H [is] a disjoint union of cliques... the least number of equivalence subgraphs needed to cover the edges of G ”), Alon’s 1986 definition (“an equivalence covering of a graph G is a family of equivalence subgraphs of G such that every edge of G is an edge of at least one member”), and Esperet–Gimbel–King’s (“an equivalence graph is a disjoint union of cliques, and the equivalence number $\text{eq}(G)$... is the minimum number of equivalence subgraphs needed to cover the edges of G ”). The writeup’s sentence “All equivalence subgraphs below may be made spanning by adding isolated vertices” is the correct and standard reconciliation between the “subgraph” and “equivalence relation on V ” formulations. **No definition gaming.**

C_n^k is used with the standard convention: $u \sim v$ iff their cyclic distance is in $[1, k]$. This matches West’s REGS page (“each vertex adjacent to the closest k vertices in each direction”). No off-by-one: $k+1$ consecutive vertices form a clique, $C_n^k = K_n$ iff $n \leq 2k+1$.

Is the refutation of the intended conjecture, or of a strawman? It is the intended conjecture. The OPG discussion is explicit that the interesting regime is n large relative to k (“even for n very large compared to k , no upper bound is known beyond trivial linear bounds of order $\Theta(k)$ ”). The writeup does *not* exploit small n : its bounds hold uniformly for every $n \geq 2k+2$, and it exhibits families with $n/k \rightarrow \infty$ and $\text{eq}/k \rightarrow 0$ (e.g. $k = 2^q - 1$, $n = 2^q \cdot 2^q$). It also correctly disposes of the degenerate regime $n \leq 2k+1$ (complete graph, $\text{eq} = 1$), so the claim covers all $n \geq 3$. The result simultaneously answers *both* halves of West’s REGS Problem 1 (“is the value unbounded for fixed k ?” — no, $\text{eq} \leq 2r+1$ uniformly in n ; “is there an upper bound sublinear in k ?” — yes, $O(\log k)$).

An author of the original problem would consider it resolved.

B.2 Step-by-step findings

$s = k+1$, $r = \lceil \log_2 s \rceil$. Claim: $r+1 \leq \text{eq}(C_n^k) \leq 2r+1$ for $n \geq 2s$, with equality $\text{eq} = r+1$ when $s \mid n$.

#	step	label	note
0	Scope: $n \leq 2k+1$ (and $n \geq 3$) gives $C_n^k = K_n$, $\text{eq} = 1$	VALID	$\max$ cyclic distance is $\lfloor n/2 \rfloor \leq k$, so the graph is complete; $\text{eq}(K_n)=1$.
1	§1 property 1: $x \in I_{h-1}(p)$, $y \in I_{h-0}(p)$ implies $x > y$	VALID	They share the first $h-1$ bits and differ at bit h with $x_{h-1} > y_{h-0}$; lower bits are irrelevant.
2	§1 property 2: $x > y$ implies both lie in a common (h, p) pair	VALID	Take h = first differing bit position of the r -bit representations; $x > y$ forces $x_{h-1} = 1$, $y_{h-0} = 0$, and p = common prefix.
3	§2: H_0 (block cliques) is an equivalence subgraph	VALID	Each block is $s = k+1$ cyclically consecutive vertices, hence a clique; blocks partition V .
4	§2: each $Q_{i,h,p}$ is a clique	VALID	Same-block pairs: within a block, distance $\leq s-1 = k$. Cross pairs $v_{i,x}$, $v_{i+1,y}$: positions $is+x$ and $(i+1)s+y$, clockwise gap $s+y-x \leq s-1 = k$ by step 1. Crucially, <i>every</i> x in the 1-part exceeds <i>every</i> y in the 0-part, which is exactly step 1 — the writeup states this correctly.
5	§2: at each level h the Q ’s are pairwise vertex-disjoint	VALID	Each $v_{i,x}$ lies in exactly one Q at level h : in $Q_{i,h,p}$ if $x_h = 1$, in $Q_{i-1,h,p}$ if $x_h = 0$, where p is x ’s length- $(h-1)$ prefix. Distinct prefixes give disjoint index sets; the left/right dichotomy is what prevents collisions. Verified by code for all $m(k+1) \leq 300$, $k \leq 24$.

#	step	label	note
6	§2: every edge is covered	VALID	An edge has short arc of length $\leq s-1 < s$, so it meets at most two consecutive blocks. Intra-block: H_0 . Inter-block $v_{\{i,x\}} v_{\{i+1,y\}}$: adjacency gives $s+y-x \leq s-1$, i.e. $x>y$, and step 2 supplies (h,p) . Includes the $m=2$ case correctly (a pair of blocks is adjacent on both sides, but each orientation is handled by a different index i).
7	§2 conclusion $\text{eq}(C_{\{ms\}}^{\wedge\{s-1\}}) \leq r+1$	VALID	Exactly $r+1$ layers. Empty/singleton Q 's (which occur when s is not a power of two) are discarded harmlessly.
8	§3 path-power cover with $r+1$ layers	VALID (minor hand-wave)	Same argument with no wraparound. The only under-argued point is “Missing vertices in the final block cause no difficulty”: when the last block is short one simply drops the out-of-range local indices, which cannot break cliqueness or disjointness and cannot orphan an edge. One line of prose; I verified it computationally for every n in $[2s, 300]$, $k \leq 24$.
9	§3 wrap cover with r layers	VALID	$A = \{n-s..n-1\}$, $B = \{0..s-1\}$ are disjoint since $n \geq 2s$, both cliques. For $a_x = n-s+x$, $b_y = y$, the wrap gap is $s-x+y$, which is $\leq k$ iff $x>y$; and every wrapping edge indeed has both ends in $A \cup B$ (its endpoints are within $s-1$ of the cut). Same disjointness argument, only one block pair involved.
10	§3 conclusion $\text{eq} \leq (r+1)+r = 2r+1$	VALID	Layers of the two covers are simply concatenated. Verified for 7390 parameter pairs.
11	§4: labels give $M_{\{uv\}}=0$ iff $u=v$ or uv in E	VALID	Components of a layer are cliques of G , so $c_1(u)=c_1(v)$ with $u \neq v$ forces uv in E ; conversely every edge is covered by some layer. Requires layers to be spanning, which the writeup arranged.
12	§4 rank bound $\text{rank } M \leq 2^t$	VALID	$\text{prod}_1(c_1(u)-c_1(v)) = \sum_{\{S \subseteq [t]\}} (-1)^{ t-S } a_S(u) a_{[t] \setminus S}(v)$, a sum of 2^t outer products. Re-derived; the sign convention is right.
13	§4 triangular minor $N_{\{ij\}} = M_{\{i, (s+j) \bmod n\}}$, $0 \leq i, j < 2s$	VALID	Rows $0..2s-1$ and columns $s..3s-1 \pmod n$ are each distinct because $2s \leq n$. For $i>j$: $ s+j-i \leq s-1$ so the two vertices are equal or adjacent, giving $N_{\{ij\}}=0$. On the diagonal the offset is exactly s , with cyclic distance $\min(s, n-s) = s > k$ since $n \geq 2s$, so $N_{\{ii\}} \neq 0$. Hence $\text{rank } N = 2s$.
14	§4 conclusion $t \geq \text{ceil}(\log_2(2s)) = r+1$	VALID	$2s \leq \text{rank } N \leq \text{rank } M \leq 2^t$; $\text{ceil}(\log_2 2s) = 1 + \text{ceil}(\log_2 s)$. Verified numerically (see below). Not new: this is Alon 1986, Theorem 1.1, instantiated at the same pair of vertex sequences (see Reference check).
15	Combination: $\text{eq}(C_{\{ms\}}^{\wedge\{s-1\}}) = r+1$ for $m \geq 2$; $\text{eq} = \Theta(\log(k+1))$ for all $n \geq 2k+2$	VALID	$r+1 \leq \text{eq} \leq 2r+1$ with $r = \text{ceil}(\log_2(k+1))$ is $\Theta(\log(k+1))$ including at $k=1$ ($2 \leq \text{eq} \leq 3$).
16	Verdict-block claim “disproves the $\Omega(k)$ conjecture”	VALID	$\text{eq}/k \rightarrow 0$ along $k=2^q-1$, $n=m \cdot 2^q$, with $n/k \rightarrow \text{infinity}$.

No step is labelled GAP or ERROR. The only prose weaknesses are step 8's one-line dismissal of the short last block and the tacit use of

“empty sets are omitted”; both are genuinely routine and were checked exhaustively by code.

B.3 Reference check

The writeup cites nothing and says so (“No unproved conjecture or external theorem is used”), so the reference check here is a *prior-art* check. Sources fetched:

1. **OpenProblemGarden entry** (http://www.openproblemgarden.org/op/covering_powers_of_cycles_with_equivalence_subgraphs) — fetched. Conjecture and discussion are quoted accurately in `prompt.md`; no comments have ever been posted on the entry. Its definition of eq matches the writeup’s.

2. **D. West, “Equivalence covering of cycle-powers” (REGS 2010/2011)** (<https://dwest.web.illinois.edu/regs/eqcov.html>) — fetched in raw form. This is the primary source for the problem (“Originators: L. Esperet, J. Gimbel, A. King, presented by Andrew King”). It states verbatim: > “Any $k+1$ consecutive vertices form a clique, and $\text{eq}(C_n^k)=k+1$ when $k+1 >$ divides n . Always $2k$ is an upper bound. Although no sublinear upper bound $>$ construction is known, also no good general lower bound is known.” and Problem 1: “Determine the asymptotics of $\text{eq}(C_n^k)$. In particular, is the value unbounded for fixed k ? Is there an upper bound that is sublinear in k ?”

The asserted equality $\text{eq}(C_n^k) = k+1$ when $(k+1) \mid n$ is false. My exhaustive computation gives $\text{eq}(C_8^3) = 3 \neq 4$, $\text{eq}(C_{12}^3) = 3 \neq 4$, $\text{eq}(C_{10}^4) = 4 \neq 5$. It is refuted by a 1986 corollary cited on that very page (item 3 below). This is almost certainly the origin of the OPG conjecture, and it means the problem was “wide open” partly on the strength of an error.

3. **N. Alon, “Covering graphs by the minimum number of equivalence relations”, *Combinatorica* 6 (1986) 201–206** — full text retrieved and read (<https://web.math.princeton.edu/~nalon/PDFS/Publications/...pdf>, extracted with `pdftotext`). Two items are directly load-bearing for this review:

- **Theorem 1.1** (verbatim): “Let $G=(V,E)$ be a graph and suppose $U=(u_1, \dots, u_s)$, $W=(w_1, \dots, w_s)$ are two (not necessarily disjoint) sequences of vertices. If $u_i w_i \notin E$ for all $1 \leq i \leq s$ and for all $1 \leq i < j \leq s$ either $u_i = w_j$ or $u_i w_j \in E$, then $\text{eq}(G) \geq \log_2 s$.” Proved by exterior algebra. **The writeup’s §4 is exactly this theorem**, with a (weaker but sufficient) outer-product/rank proof in place of Alon’s exterior-algebra proof, applied to $u_i = 2s-1-i$, $w_i = (s + \lfloor 2s-1-i \rfloor) \bmod n$, $i=0 \dots 2s-1$. I verified by code that Alon’s hypotheses hold verbatim for that pair of sequences for every tested $n = m(k+1)$, $m \geq 2$, $k \leq 15$, giving $\text{eq} \geq \log_2(2(k+1))$. So §4 is a rediscovery, not new mathematics. (The writeup does not claim otherwise; it explicitly says it has not checked the literature.)
- **Corollary 1.2** (verbatim): “Let T_n denote the complement of a matching of $n/2$ edges. Then $\text{eq}(T_n) = \lceil \log_2 n \rceil$ for all even $n \geq 2$.” Since $C_{2k+2}^k = T_{2k+2}$, this says $\text{eq}(C_{2k+2}^k) = \lceil \log_2(2k+2) \rceil = r+1$, i.e. **the $n = 2k+2$ case of the writeup’s theorem has been known since 1986** — and it already contradicts West’s $\text{eq} = k+1$ claim ($k=3: 3 \neq 4$). Alon’s Section 4 “Concluding remarks” contains no statement about cycle powers.

4. **L. Esperet, J. Gimbel, A. King, “Covering line graphs with equivalence relations”, *Discrete Appl. Math.* 158 (2010) 1902–1907** — extended abstract retrieved (<https://imada.sdu.dk/u/btoft/GT2009/esperet.pdf>) and text-extracted. Main theorem confirmed as $(1/3) \log_2 \log_2 \chi(G) < \text{eq}(L(G)) \leq 2 \log_2 \log_2 \chi(G) + 2$. The paper is about line graphs and the orientation covering number $\sigma(G)$; **it contains nothing about C_n^k** , so it is not prior art for this claim. The writeup does not cite it.

5. **Post-2011 literature.** Searches for a sublinear upper bound or any resolution of West’s Problem 1 returned nothing: arXiv:2010.04450 (“A note on the orientation covering number”) proves $\sigma(G) = \sigma(K_{\chi(G)})$ and does not touch cycle powers; arXiv:1608.07723 (Javadi-Hajebi) is about the *edge clique cover* number $\text{cc}(C_n^k)=n$, a different and larger parameter. This agrees with the catalog page’s own 2026 review.

Conclusion of the reference check: the *lower bound* and the special case $n = 2k+2$ are classical (Alon 1986). The *upper bound* for general n — in particular for $n \gg k$, the regime the conjecture is about — is not in the literature I could find, and it is what actually refutes the $\Omega(k)$ conjecture.

B.4 Computational check

Scripts in `verification_astra/scripts/covering_powers_of_cycles_with_equivalence_subgraphs/` (run with system python3 3.x + numpy 1.26.4; networkx not needed):

- `construction_check.py` — re-implements the §2 and §3 constructions from the writeup’s own definitions and independently verifies, against a freshly built edge set of C_n^k , that (a) every part is a clique, (b) the parts of a layer are pairwise disjoint, (c) the layers cover every edge.
- `brute_force_eq.py` — computes $\text{eq}(C_n^k)$ **exactly** by enumerating *all* partitions of V into cliques of G and solving the resulting minimum set cover exactly (maximal-set reduction + iterative deepening branching on a least-coverable edge).
- `curated_cases.py` — exact eq on a curated list, cross-checked against the writeup’s formula and against West’s $\text{eq} = k+1$ claim.
- `rank_lower_bound.py` — builds the §4 matrix M from an actual cover with random real labels and checks the zero pattern, $\text{rank } M \leq 2^t$, the triangularity of the $2s \times 2s$ minor N and $\text{rank } N = 2s$; also tests Alon’s Theorem 1.1 hypothesis on the same sequences.
- `witnesses.py` — prints explicit small covers.

Result 1 — constructions valid (7390 parameter pairs, 0 failures). §2 verified for every $n = m(k+1) \leq 300$ with $m \geq 2$, $k \leq 24$, always using exactly $r+1$ layers; §3 verified for every n in $[2k+2, 300]$, $k \leq 24$, never exceeding $2r+1$ layers.

Result 2 — exact eq by exhaustive search. ($r+1$ = writeup’s lower bound / exact value in the divisible case; $2r+1$ = writeup’s upper bound.)

n	k	exact eq	r+1	2r+1	k+1	(k+1) n	
4	1	2	2	3	2	yes	ok
5	1	3	2	3	2	no	ok
6	1	2	2	3	2	yes	ok
7	1	3	2	3	2	no	ok
8	1	2	2	3	2	yes	ok
9	1	3	2	3	2	no	ok
6	2	3	3	5	3	yes	ok
7	2	4	3	5	3	no	ok
8	2	3	3	5	3	no	ok
9	2	3	3	5	3	yes	ok
10	2	3	3	5	3	no	ok
11	2	3	3	5	3	no	ok
12	2	3	3	5	3	yes	ok
8	3	3	3	5	4	yes	West's eq=k+1 re- futed
9	3	4	3	5	4	no	ok
10	3	4	3	5	4	no	ok
11	3	4	3	5	4	no	ok
12	3	3	3	5	4	yes	West's eq=k+1 re- futed
10	4	4	4	7	5	yes	West's eq=k+1 re- futed
11	4	4	4	7	5	no	ok

Every value lies in $[r+1, 2r+1]$; **every divisible case equals $r+1$ exactly**, as the writeup claims, and never equals $k+1$ when $k \geq 3$. Cases with $n \geq 12$ and $k \geq 4$ exceeded the enumeration budget (the number of clique partitions explodes) and were not completed; they are not needed, since the upper bound is constructive and the lower bound is Alon's theorem.

Note that $n=12, k=3$ is an $m=3$ case — it is *not* covered by Alon's Corollary 1.2, so the new regime of the writeup's theorem is independently confirmed by exhaustive search.

Result 3 — explicit witnesses (all machine-verified to be equivalence covers):

C_{8^3} ($k+1 = 4, 2k = 6$), 3 layers:
 $\{0,1,2,3\} \{4,5,6,7\}$
 $\{2,3,4,5\} \{0,1,6,7\}$
 $\{1,4\} \{3,6\} \{0,5\} \{2,7\}$
 C_{12^3} ($k+1 = 4, 2k = 6$), 3 layers:
 $\{0,1,2,3\} \{4,5,6,7\} \{8,9,10,11\}$
 $\{2,3,4,5\} \{6,7,8,9\} \{0,1,10,11\}$
 $\{1,4\} \{3,6\} \{5,8\} \{7,10\} \{0,9\} \{2,11\}$
 C_{24^7} ($k+1 = 8, 2k = 14$), 4 layers:
 $\{0..7\} \{8..15\} \{16..23\}$
 $\{4..11\} \{12..19\} \{0,1,2,3,20,21,22,23\}$
 $\{2,3,8,9\} \{6,7,12,13\} \{10,11,16,17\} \{14,15,20,21\} \{0,1,18,19\} \{4,5,22,23\}$
 $\{1,8\} \{3,10\} \{5,12\} \{7,14\} \{9,16\} \{11,18\} \{13,20\} \{15,22\} \{0,17\} \{2,19\} \{4,21\} \{6,23\}$

Result 4 — §4 verified numerically. For all tested (n,k) with $n=m(k+1), m \in \{2,3\}, k \leq 15$: the zero pattern of M is exactly " $u=v$ or uv in E "; the $2s \times 2s$ minor N is triangular with nonzero diagonal and has rank exactly $2s$; $\text{rank } M \leq 2^t$; and $2s \leq 2^t$ in every case (e.g. $n=32, k=15: t=5, \text{rank } N = 2s = 32 = 2^5$, tight). Alon's Theorem 1.1 hypothesis holds verbatim on the (order-reversed) same sequences.

B.5 Caveats

- Prior art, lower bound.** §4 is Alon (1986) Theorem 1.1 rediscovered, with a rank proof instead of exterior algebra. The writeup makes no novelty claim, but a submitted version must cite it.
- Prior art, special case.** $\text{eq}(C_{2k+2}^k) = \text{ceil}(\log_2(2k+2))$ is Alon (1986) Corollary 1.2. The $n = 2k+2$ slice of the "new" theorem is 40 years old; the novelty is the extension to $m \geq 3$ blocks (and hence to $n \gg k$).
- The contradicted claim in the source is itself erroneous.** West's REGS page asserts $\text{eq}(C_n^k) = k+1$ when $(k+1) \mid n$, which is false and is contradicted by Alon's corollary cited on the same page. The problem's "wide open" status rests partly on that slip, which lowers the novelty bar considerably — the disproof is a natural m -block generalization of a 1986 construction. The verdict block's `would_publish: true` is optimistic on that ground (the mathematics is right; the contribution is modest and needs the 1986 context).
- Not determined:** the exact value of $\text{eq}(C_n^k)$ when $(k+1) \nmid n$. The writeup says so. My data show the gap is real ($\text{eq}(C_7^2)=4 > r+1 = 3, \text{eq}(C_9^3)=\text{eq}(C_{10}^3)=\text{eq}(C_{11}^3)=4 > 3$), so the non-divisible case is genuinely harder and $r+1$ is not the answer there. The $2r+1$ upper bound is not claimed tight and is not tight ($\text{eq}(C_7^2) = 4 < 5$).

5. **Edge cases.** $k=0$ is excluded (C_n^0 is edgeless, $eq = 0$); the writeup assumes $k \geq 1$. $n \leq 2k+1$ is handled (K_n , $eq=1$), with the implicit assumption $n \geq 3$. $n = 2k+2$ exactly is fine ($m=2$): the two blocks are cyclically adjacent on both sides, and the construction handles each orientation with a different index i ; I checked this case explicitly. The s not a power of two case leaves some $I_{h+1}(p)$ empty, producing singleton “cliques” that are discarded — harmless, and the layer count can only drop below $r+1$.
6. **Convention dependence.** The result would change if “equivalence subgraph” were required to be *induced* or to consist of *maximal* cliques of G — but no source uses such a definition, and the writeup’s §2 layers do use non-maximal and non-consecutive cliques (e.g. $\{1,4\}$ in C_8^3), which the writeup flags honestly (“Some of these cliques have gaps in the cyclic order. That is allowed”).
7. **Uniformity in n .** The claim is $eq = \Theta(\log(k+1))$ uniformly over $n \geq 2k+2$; the hidden constants are 1 and 2 on r , so the statement is genuinely uniform. No hidden dependence on n slipped in.

B.6 Referee summary

I attacked this writeup expecting the usual failure modes and found none. The construction is a clean binary “left/right boundary” layering: H_0 is the partition into $k+1$ -blocks, and level h merges the vertices whose h -th bit is 1 in block i with those whose h -th bit is 0 in block $i+1$; the cliqueness relies on the fact that every index in the 1-part exceeds every index in the 0-part, and the disjointness on the fact that a vertex crosses either its right or its left boundary but never both. Both facts are stated correctly and I verified all three required properties by code for 7390 parameter pairs ($k \leq 24$, $n \leq 300$, zero failures). The lower bound is a correct rank argument, but it is Alon’s 1986 Theorem 1.1 rediscovered, and its $n = 2k+2$ instance is Alon’s Corollary 1.2, so the exact formula is only new for $m \geq 3$ blocks. Exhaustive computation of $eq(C_n^k)$ over *all* clique partitions confirms the formula in every divisible case I could complete ($eq(C_6^2)=eq(C_9^2)=eq(C_{12}^2)=3$, $eq(C_8^3)=eq(C_{12}^3)=3$, $eq(C_{10}^4)=4$) and confirms $r+1 \leq eq \leq 2r+1$ in every non-divisible case. The conjecture $eq(C_n^k) = \Omega(k)$ is therefore genuinely false, in the intended regime $n \gg k$. The one uncomfortable discovery is that the problem’s own source page (West’s REGS page, the origin of the OPG entry) asserts $eq(C_n^k) = k+1$ when $(k+1) \mid n$, which my brute force refutes at $n=8, k=3$ and which Alon’s Corollary 1.2 — cited on that same page — already refuted in 1986. The mathematics in the writeup is sound and the disproof stands; its novelty is real but narrower than the verdict block suggests, and any publishable version must cite Alon (1986) for the lower bound and for the $n=2k+2$ case.

References

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- [2] D. B. West, *Equivalence covering of cycle-powers*, REGS problem page (2010/2011), <https://dwest.web.illinois.edu/regs/eqcov.html>.
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